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ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

MARKING SCHEME

NOVEMBER 2014

CHEMISTRY 9189/2

1 (a) $Kp = \frac{P^2_{NH3}}{P^3_{H_2}.P_{N_2}}$; reject square brackets. $\frac{P_{NH_3}{}^2}{P^3_{H_2}.P_{N_2}}$ or $\frac{(P_{NH_3})^2}{(P_{H_2})^3.P_{N_2}}$ [1]

(b)

$$\begin{array}{cccc} N_2 & + & 3H_2 & \rightleftharpoons 2NH_3 \\ 2 & & 2 & 0 \\ 2-x & & 2-3x & 2x \end{array}$$

$x = 0.2$

at equilibrium moles of N_2 = 1.8 mols; [1]

moles of H_2 = 1.4 moles; [1]

moles of NH_3 = 0.4 moles; [1]

mark for the answer only.

(c) $P_{N2} = \frac{1.8}{3.6}P_T$

$P_{H2} = \frac{1.4}{3.6}P_T$

$PN_{H3} = \frac{0.4}{3.6}P_T$;

*ecf from (b) to (c)***

* No error cf on the same part of qn.

[1]

$Kp = \frac{P^2_{NH_3}}{P_{N2}.P^3_{H_2}}$ ~~[1]~~

$= \left(\frac{0.4}{3.6p_T}\right)^2 / \left(\frac{1.4}{3.6p_T}\right)^3 \left(\frac{1.8}{3.6p_T}\right)$; [1]

$40.7 = \frac{\cancel{0.16}\ 0.418}{P_T{}^2}$; ~~0.02003~~ 0.~~184~~ 0.~~108~~ ~~[1]~~

P_T = ~~0.063 atm~~; 0.102 atm [1]

Total (7)

2 (a) The potential difference ~~(or emf)~~ of half cell measured relative to a standard hydrogen electrode; under std cdns. [1]

(b) (i) temperature of 298 K; / AW 25°C [1]

pressure of 1 atm; [1]

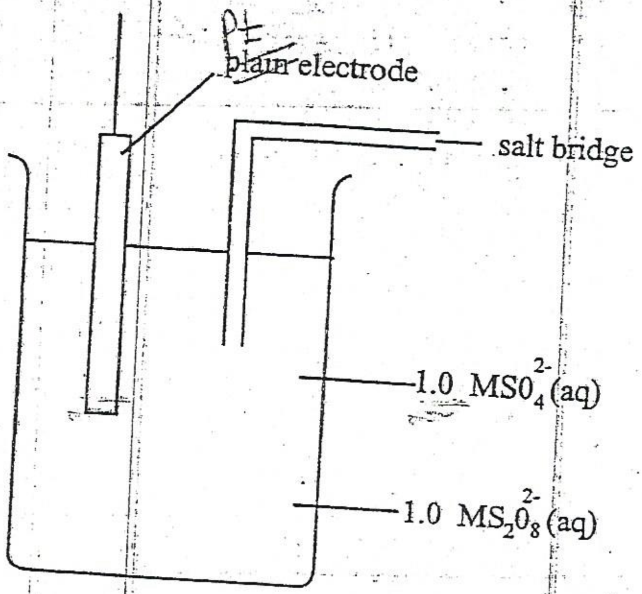
solutions with concentrations of 1.0 mol/dm³; AW 1M [1]

(ii) $S_2O_8^{2-} + 2Ag \rightarrow 2SO_4^{2-} + 2Ag^+$ (Reject ⇌) [1]

(iii) $E^\theta = 2.01 - 0.80$

$= (+) 1.21\ V;$ [1]

~~(iii)~~ (iv)



diagrams; [1]

labels; (all labels) [1]

(c) ~~E^θcell increases;~~ /AW - more positive. [1]

/ fwd rxn favoured. (AW)

$[Ag^+]$ is reduced by NH_3 due to formation of $Ag(NH_3)_2^+$ [1]

equilibrium position shifts to the right; [1]

Total 11

3 (a) (i) combustion of sulphur containing fuels/fossil fuels; /AW coal / sulphur containing ores [1]

1

(ii) causes formation of acid rain;

~~leading to~~ corrosion of buildings/

destruction of vegetation/death of

aquatic organisms

2

(iii) reducing agent/Retard /slows down the oxidation of foodstuffs/forms a weakly acidic solution which kills bacteria;

1

(b) (i) the forward reaction is exothermic and is favoured by a low temperature; [1]

(ii) high pressure requires thicker pipes that are expensive; [1]

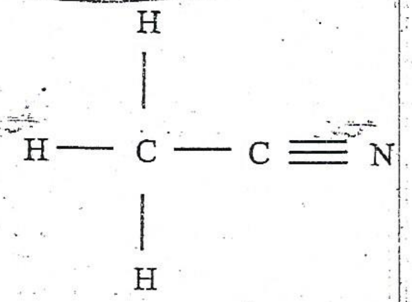
(iii) Excess air increases the ~~concentration~~ amount of oxygen and promote the forward reaction; [1]

(iv) increase rate of reaction [1]

4 (a) (i) Step I KCN; (1) reflux / HCN reaction.
ethanol at rtp/ (1) alkaline NaOH. [2]

Step II $LiAlH_4$; (1) / $NaBH_4$
Ether followed by water (1) aqueous
~~ethanol reflux~~
heat (A) [4]

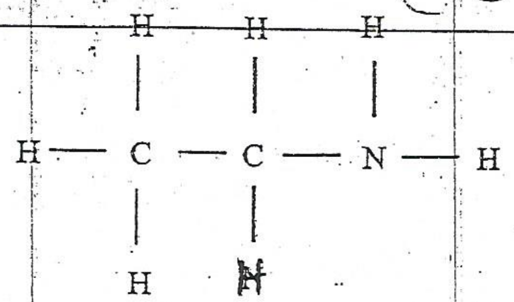
(ii)

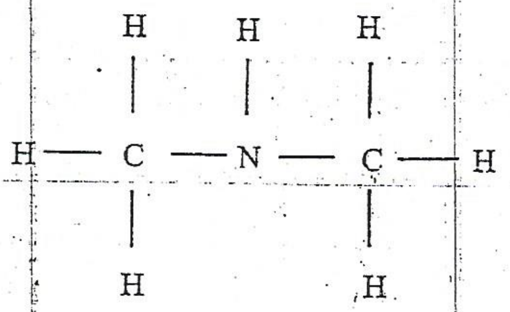


[1]

(b) (i) $CH_3CH_2NH_2$;
for both correct [1]

$(CH_3)_2NH$ [1] A
A $CH_3CH_2NH_2$





[1]

18
14
32

16
14

F 70.

(ii) two methyl groups have greater positive inductive effect than one ethyl group/AW; [1] Any

the +ve cation formed more stabilised in dimethylamine than in ethylamine; [1] [2]

One pair of electrons drawn by the oxygen atom of the carbonyl by resonance;

AW One pair of election in ~~ethylamine~~ dimethylamine available for dative bond with a proton/AW;

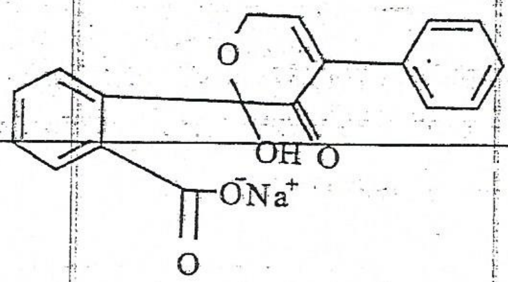
4. C (i) refrigerants; solvents;
aerosol propellants; [2]
Pesticides / Flame retardants / fire extinguishers / PVC — Any two [2]

(ii) ozone depletion;
global warming; [2]
photochemical smog — Any two [2]

5 (a) ester;
keto; reject carbonyl
alkene; [2.]
ether;
phenyl; / arene / Arenyl / benzene ring — Any ~~three~~ two

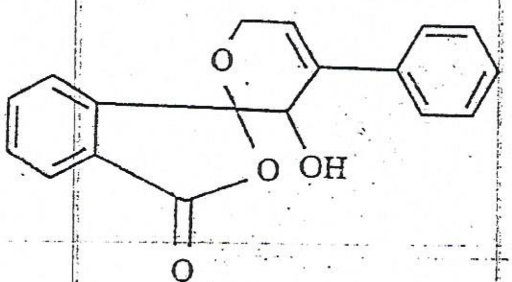
Total [12]

(b) (i)



[1]

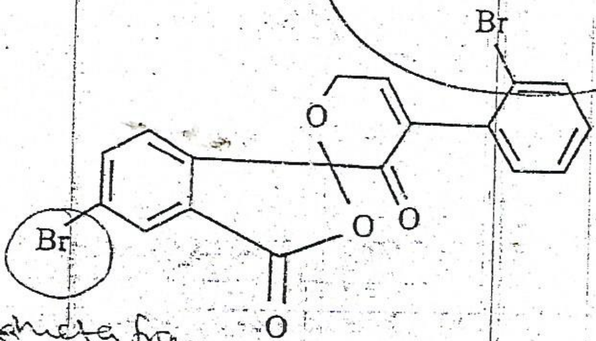
(ii)



[1]

1345

(iii)



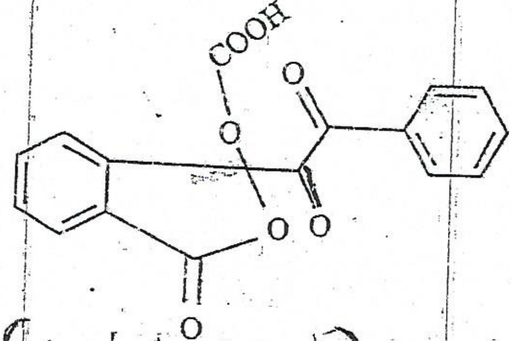
[1]

*meta from ester

Monosubstituted benzene ring with benzene on ortho or para position;

Bromine on meta position;

(iv)



[1]

(Acid hydrolysis)

(c) (i) nylon-6,6 is hydrolysed by the acid to its monomers; [1] [1]

(ii) hydrocarbon sections of the nylon chain represent a higher proportion of the total length of the polymer; (hydrophobic) [1]

hence less of the molecule is available for water absorption by hydrogen bonding; [1] [1]

(iii) has a high nitrogen content; (1) [1]

$10 x 2

~~$1 x 0,50~~

Total 10

Terylne → hydrophilic).